



vc-7434

**Integrated M. C. A. (Sem. III)
Examination, December - 2023**

IMCA-304

Statistics and Probability

Time Allowed : Three Hours

Maximum Marks : 70

Minimum Pass Marks : 28

***Note : This question paper contains three
sections as under :***

SECTION - B

UNIT - I

- 2 What is the chance that a leap year, selected at random will contain 53 Sundays?

OR

A and B throw alternatively with a pair of ordinary dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, show that his chance of winning is $\frac{30}{61}$.

UNIT - II

- 3 Derive of Binomial Distribution as the limiting case of Hypergeometric Distribution.

OR

In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the parameters of the distribution.

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[Contd...

UNIT - III

- 4 Find the MLE for the parameter λ of Poisson distribution.

OR

Show that in a normal distribution $\bar{x}$ is a minimum variance estimator of μ .

UNIT - IV

- 5 Two samples of size 9 and 8 gave the sums of squares of deviations from their respective means equal to 160 and 91 respectively. Can they be regarded as drawn from the same normal population?

OR

Write any five use of Z-test.

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[Contd...